

REPORT

Title

Prolog Program for Planets Database

Aim

To create a **Planets database in Prolog** and retrieve planet information using logical queries.

Problem Statement

Create a Prolog database containing details of planets such as their name, position from the Sun, and type. The program should allow users to retrieve the stored information.

Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp4.pl`)

Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp4.pl`.
3. Enter the planet database facts.
4. Save the file.
5. Load the program using `[exp4]..`
6. Enter queries to retrieve planet details.
7. Observe the output.

Program

```
planet(mercury, 1, terrestrial).  
planet(venus, 2, terrestrial).  
planet(earth, 3, terrestrial).  
planet(mars, 4, terrestrial).  
planet(jupiter, 5, gas_giant).  
planet(saturn, 6, gas_giant).  
planet(uranus, 7, ice_giant).  
planet(neptune, 8, ice_giant).
```

Sample Output

Query:

```
?- [exp4].  
true.
```

```
?- planet(earth, Position, Type).
```

Output:

```
Position = 3,  
Type = terrestrial.
```

Another query:

```
?- planet(Planet, 5, Type).
```

Output:

```
Planet = jupiter,  
Type = gas_giant.
```

To display all planets:

```
?- planet(Planet, Position, Type).
```

Conclusion

Thus, the **Planets database** was successfully created in Prolog. The program can retrieve planet details such as **name, position, and type** using simple logical queries.